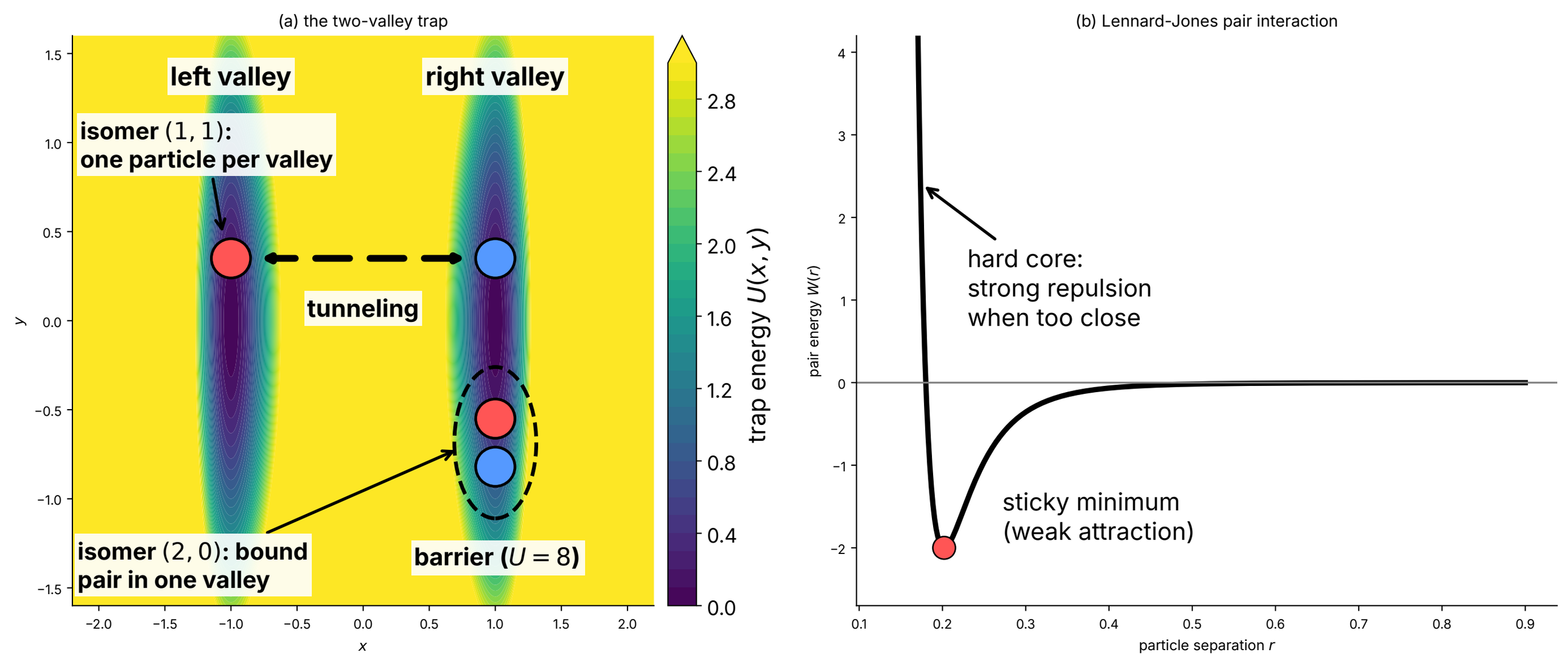

(a) the two-valley trap
left valley
right valley
isomer (1, 1):
one particle per valley
tunneling
isomer (2, 0): bound
pair in one valley
barrier (U = 8)
trap energy U(x, y)
x
y
(b) Lennard-Jones pair interaction
hard core:
strong repulsion
when too close
sticky minimum
(weak attraction)
pair energy W(r)
particle separation r